

TASK -1

FOUNDATIONS OF CYBERSECURITY

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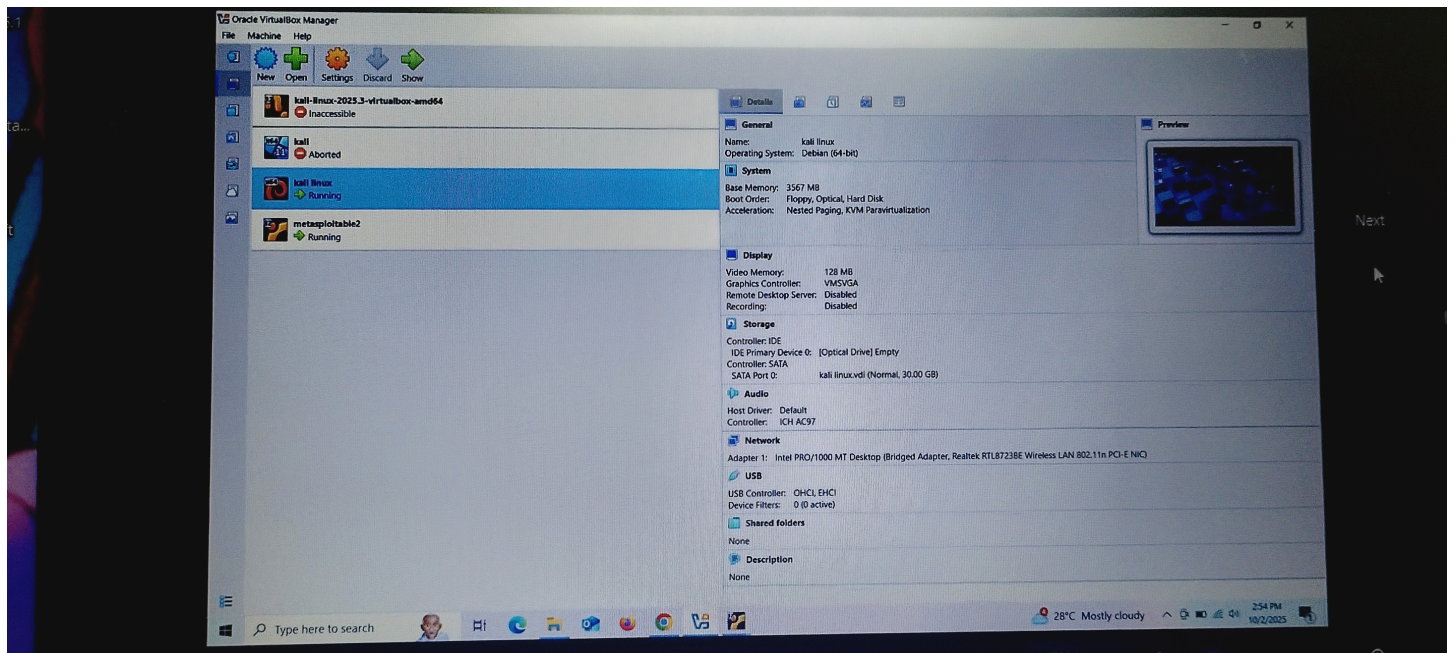
DATE : 12/11/2025

● OBJECTIVE

To build a cybersecurity lab and learn basic Linux, networking, cryptography, and tool usage.

● LAB SETUP

I installed Kali Linux (attacker) and Metasploitable 2 (target) in VirtualBox and configured Host-Only network.



● LINUX FUNDAMENTALS

-PWD: Shows the current working directory

-LS: lists files and directories

-IFCONFIG: shows the ip addresses

-TOUCH: creates a new file

-APT UPDATE: updates the package list


```
plain.txt nikhitha@kali: ~
Session Actions Edit View Help
(nikhitha@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:fe81:3190 prefixlen 64 scopeid 0<link>
    ether 08:00:27:81:31:90 txqueuelen 1000 (Ethernet)
    RX packets 13 bytes 11174 (10.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 38 bytes 11104 (10.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(nikhitha@kali)-[~]
$ echo "this is a secret message for task 1 - openssl demo">plain.txt

(nikhitha@kali)-[~]
$ ls -l plain.txt encrypted.bin
-rw-r--r-- 1 nikhitha kali 33 2024-10-27 10:00 encrypted.bin
-rw-r--r-- 1 nikhitha kali 33 2024-10-27 10:00 plain.txt
```

```
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(nikhitha@kali)-[~]
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-rw-r--r-- 1 nikhitha kali 33 2024-10-27 10:00 encrypted.bin
-rw-r--r-- 1 nikhitha kali 33 2024-10-27 10:00 plain.txt

(nikhitha@kali)-[~]
$ openssl enc -d -aes-256-cbc -pbkdf2 -in encrypted.bin -out decrypted.txt
enter AES-256-CBC decryption password:

(nikhitha@kali)-[~]
$ ls
decrypted.txt  Documents  encrypted.bin  Pictures  Public  Templates
Desktop       Downloads  Music          plain.txt  sample  Videos

(nikhitha@kali)-[~]
$ pwd
/home/nikhitha

(nikhitha@kali)-[~]
$ chmod
chmod: missing operand
Try 'chmod --help' for more information.
```

● OPENSSL DEMO(CRYPTOGRAPHY)

ENCRYPTED COMMAND: echo "This is a secret message for Task 1 - OpenSSL demo" > plain.txt

DECRYPTED COMMAND: openssl enc -d -aes-256-cbc -pbkdf2 -in encrypted.bin -out decrypted.txt

OUTPUT:

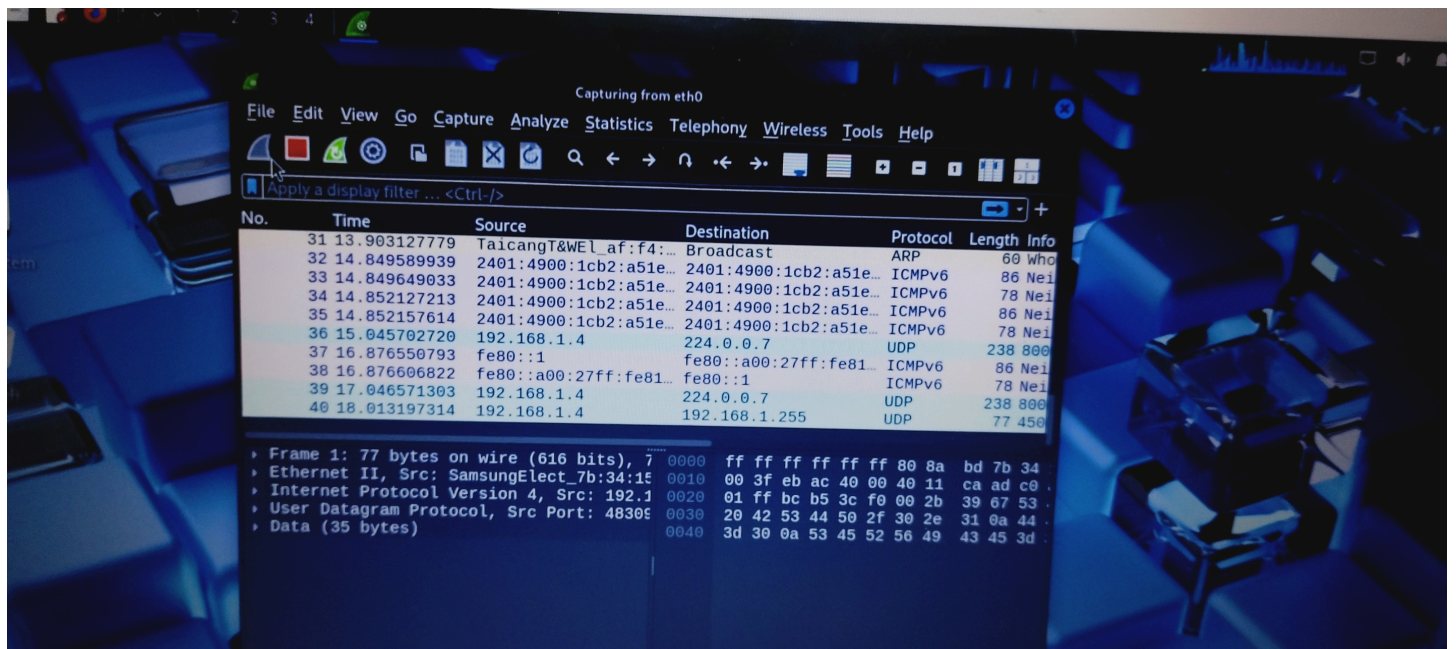

```
plain.txt
nikhitha@kali: ~
Session Actions Edit View Help
(nikhitha@kali)-[~]
$ echo "this is a secret message for task 1 - openssl demo">plain.txt
(nikhitha@kali)-[~]
$ openssl enc -aes-256-cbc -pbkdf2 -salt -in plain.txt -out encrypted.bin
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
(nikhitha@kali)-[~]
$ ls -l plain.txt encrypted.bin
-rw-r--r-- 1 nikhitha nikhitha 64 2023-10-10 10:10 encrypted.bin
-rw-r--r-- 1 nikhitha nikhitha 28 2023-10-10 10:10 plain.txt
(nikhitha@kali)-[~]
$ openssl enc -d -aes-256-cbc -pbkdf2 -in encrypted.bin -out decrypted.txt
enter AES-256-CBC decryption password:
(nikhitha@kali)-[~]
$ cat decrypted.txt
this is a secret message for task 1 - openssl demo
(nikhitha@kali)-[~]
$
```

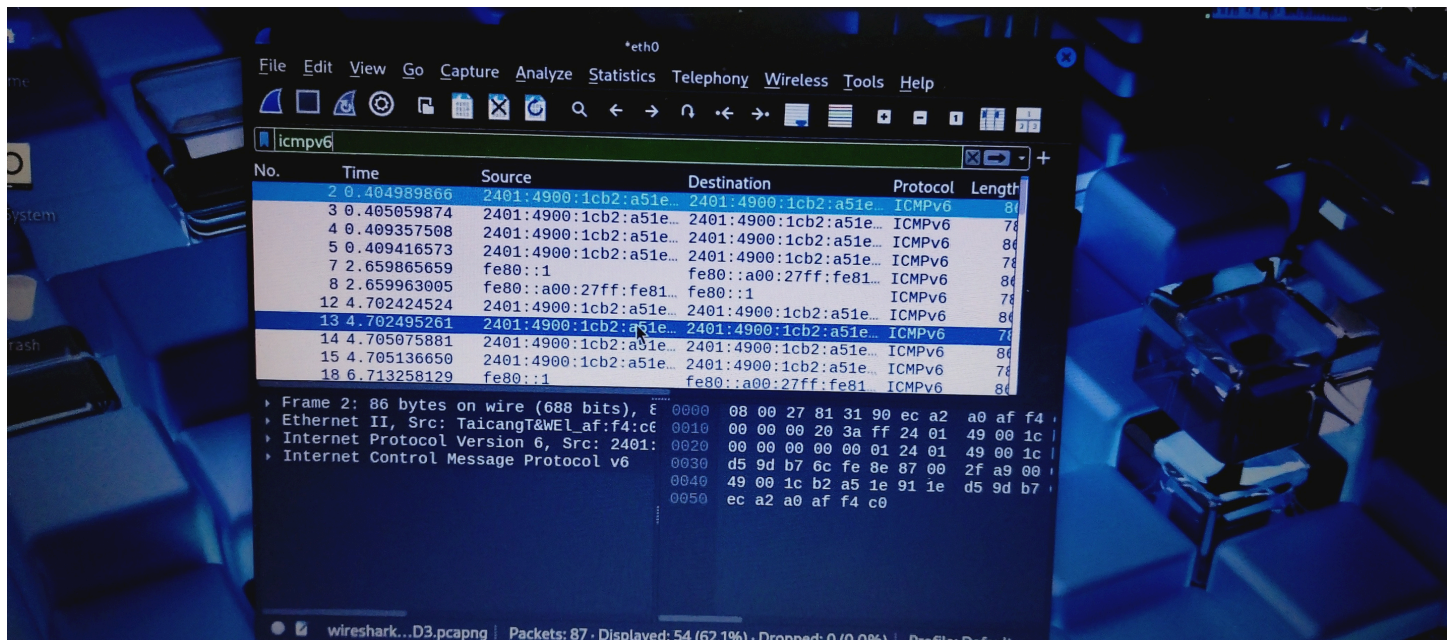
● TOOL FAMILIARIZATION

Wireshark — Packet capture & analysis

What it does: Captures network traffic and displays packet-level details (source/destination IP, protocols, payloads). Useful for spotting unencrypted credentials and protocol problems.

How I used it: Captured traffic while ping the target VM and filtered HTTP traffic.



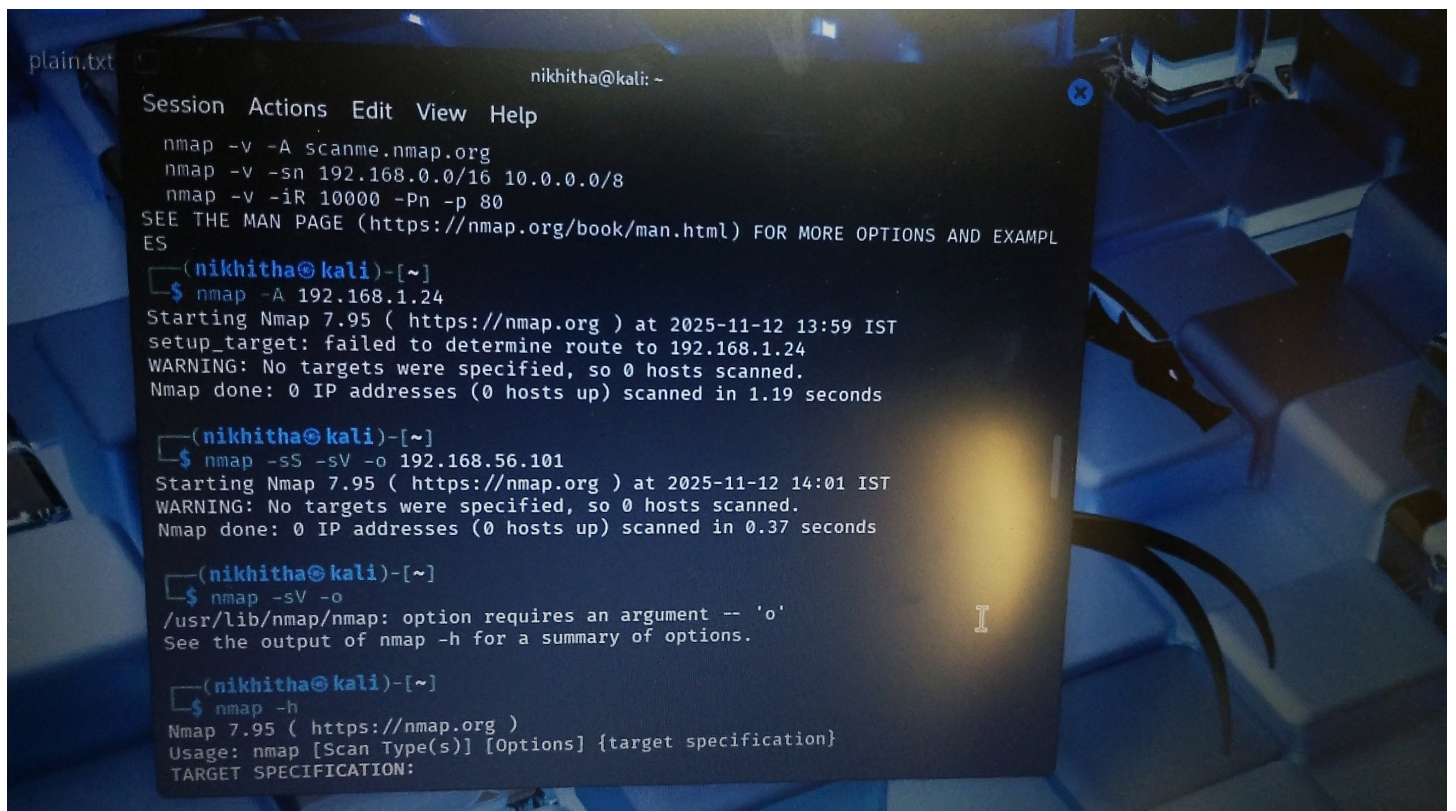


Nmap — Network & port scanner

What it does: Discovers hosts, open ports, services, and OS details on a network.

How I used it: Ran service/version and OS detection on the target VM.

Sample command: `nmap -sS -sV -O 192.168.56.101`



Netcat (nc) — Network debugging / simple server & client

What it does: Read/write data across network connections; useful for quick listeners/reverse shells in labs.

How I used it: Listened on a test port and sent a small message from the target.

Sample command: `nc -lvp 4444` (listener)


```
nc -l 4444
nc -lvp 4444
^C

(nikhitha@kali)-[~]
$ sudo nc -lvp 4444
[sudo] password for nikhitha:
nc: invalid option -- '1'
nc -h for help

(nikhitha@kali)-[~]
$ sudo nc -lvp 4444
listening on [any] 4444 ...
nc 192.168.56.101
56789
connect to [192.168.56.1] from 192.168.56.101 56789
```

OpenSSL — Cryptography toolkit

What it does: Encrypt, decrypt, sign, and verify files; generate keys and certificates.

How I used it: Performed AES-256 symmetric encryption and RSA public/private encryption.

Sample command:

`openssl enc -aes-256-cbc -pbkdf2 -salt -in plain.txt -out encrypted.bin`

```
Session Actions Edit View Help

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encrypted.bin
plain.txt

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• WHAT I LEARNED

During Task 1 I successfully built a local cybersecurity lab and practiced essential Linux and networking commands. I learned to install and operate Kali Linux and a vulnerable target VM, configure a host-only network, and verify connectivity using ping. I gained hands-on familiarity with important tools: Nmap for discovery and reconnaissance, Wireshark for packet capture and analysis, and OpenSSL for cryptographic operations (AES symmetric encryption and RSA key operations). Practically performing encryption and decryption helped me understand key management and the differences between symmetric and asymmetric cryptography. Overall, this task strengthened my foundational skills required for secure system administration and penetration testing.